

Score-formula speaker notes

Q&A-prep reference for the **Scoring a trajectory** slide (visible) and the **Cleanliness sub-score** slide (hidden). All numeric stats verified against `final_report/methodology/methodology.tex` and `final_report/results/results.tex`.

Table 1 — $J(\tau)$ terms

Term	Weight	What it measures	Why this term & weight	Stats support
β (base-line)	50	Anchor of the $[0, 100]$ clamp. A trajectory with no cleanliness gain and no penalties scores exactly 50.	Symmetric baseline — clean trajectories rise toward 100, penalised ones fall toward 0. Lets trajectories of different lengths sit on one scale. (<i>methodology.tex:64–71</i>)	Not a fitted weight. Monotone-recovery confirms $\text{sum}/\text{sum} \rightarrow 0$ when all terms stripped, so the score's signal comes from the weighted terms, not the baseline. (<i>Table tab:results-ablation-monotone</i>)
$W_g \cdot \Delta C(\tau)$ — cleanliness gain	+50	Endpoint cleanliness change: $\Delta C = C(s_T) - C(s_0)$. Each unit of gain adds 50 points.	Main positive term. Cedrim et al. show refactoring often leaves structural metrics unchanged or worse, so improvement must be rewarded explicitly. Paixão et al. justify treating cleanliness gain and process penalties as competing parts of one score. Sized high so a plausible single-step gain can't outweigh a broken build. (<i>methodology.tex:111–117; cedrim2017refactoring; paixao2017balancing</i>)	• Solo recovery = 0.000 on all sets (cancels — user and alternative validate against same terminal AST).• LOO recovery >1.0 (1.060 inj, 1.120 us, 1.108 agent) — it matters via clamp interaction.• Single-knob top-1 disruption = 10.7% — largest of all weights. (<i>Tables tab:results-ablation-solo, tab:results-ablation-loo, tab:results-sensitivity-knobs</i>)
$W_l \cdot \sigma_l(\tau)$ — step-savings bonus	+11	Step-savings bonus credited only to alternatives that reach the same end state in fewer steps. The user trajectory is never penalised for length.	Mirrors search-based refactoring's multi-objective treatment of operation count. Rewards efficient alternatives without punishing necessarily-long user paths. Weight equal to W_{mi} and W_{lag} , half of W_b — literature-severity ordering. (<i>methodology.tex:127–128; harman2007pareto; mohan2019manyobjective</i>)	• Solo recovery = 0.426 on user-study — largest single-term contributor .• LOO drops user-study magnitude to 0.636 — biggest leave-one-out.• LOO Kendall $\tau = 0.527$ on user-study — the clearest rank-carrier.• Single-knob top-1 disruption = 6.2%. (<i>Tables tab:results-ablation-solo, tab:results-ablation-loo, tab:results-ablation-user-loo</i>)
$W_b \cdot r_b(\tau)$ — broken build / test rate	–28	Fraction of <i>wall-clock</i> time the build or tests were broken: $r_b = b_{ms} / e_{ms}$. Time-weighted so it's insensitive to sample rate.	Broken state = behaviour preservation temporarily abandoned — the largest single penalty weight. Paixão et al. show developers give up ~25% of available metric improvement to avoid disruptive intermediates. Counter-evidence acknowledged honestly: Gautam et al.'s RefactorBench shows rejecting every broken intermediate can hurt agent performance on multi-file edits. (<i>methodology.tex:114–117; paixao2017balancing; gautam2025refactorbench</i>)	• Solo recovery = 0.302 (inj), 0.225 (us) — 2nd-largest on injection set.• LOO Kendall $\tau = 0.192$ on injection — strong rank-carrier in controlled setup.• Single-knob top-1 disruption = 5.3%. (<i>Tables tab:results-ablation-solo, tab:results-ablation-loo</i>)
$W_{st} \cdot r_{st}(\tau)$ — skipped-test rate	–14	Laplace-smoothed event-rate of 60-second refactoring windows with no test run: $r_{st} = (k+1)/(n+2)$.	Skipping tests removes the developer's main feedback signal that behaviour really has been preserved. 60-s window motivated by Murphy-Hill et al.'s finding that developers test in batches, not after every edit. Weight = $\frac{1}{2} \times W_b$: weaker evidence of process damage than an observed broken state. (<i>methodology.tex:119–120; murphyhill2009howrefactor</i>)	• Solo recovery = 0.371 on user-study — 2nd-largest, <i>bigger than broken</i> on natural traces. Report attributes this to “the more natural user traces expose test-running behaviour more clearly than the controlled injection set”.• LOO $\tau = 0.663$ on user-study.• Single-knob top-1 disruption = 6.7%. (<i>Table tab:results-ablation-solo</i>)
$W_{mi} \cdot r_{mi}(\tau)$ — manual edits the IDE could refactor	–11	Laplace-smoothed rate of manual edits accomplishing what an IDE refactoring could have done. Detected by RefactoringMiner over sliding commit windows in the shadow repo.	Hand-edits bypass IDE precondition checks, making subtle behavioural slips easier. Negara/Vakilian confirm developers often refactor manually even when an automated equivalent exists. Weight not fitted — no published study links fault rate to manual-vs-IDE; follows literature-severity ordering. (<i>methodology.tex:122–125; mens-tourwe; negara2013; vakilian2012</i>)	• Solo recovery = 0.261 (inj), 0.170 (us), 1.000 (agent) — 41/42 agent DPs are Manual-Refactor; treat with care, it's a detector-shape artefact on agents, not a general property.• LOO $\tau = 0.588$ on user-study — 2nd-largest rank effect after length.• Single-knob top-1 disruption = 7.6%. (<i>Tables tab:results-ablation-solo, tab:results-ablation-user-loo</i>)
$W_{lag} \cdot l(\tau)$ — intermediate cleanliness lag (degradation)	–11	$l(\tau) \in [0, 1]$: per-checkpoint running mean of $\max(0, C(s_T) - C(s_i))$. Positive while the trajectory sits below its final cleanliness — penalises time in degraded intermediate states.	Endpoint gain alone can't tell apart two trajectories with the same endpoints but different intermediate quality. Adds a delay cost — Cunningham's technical-debt metaphor and Kruchten et al.'s formalisation both treat degraded code as more costly the longer it remains. Cedrim et al. show intermediate states can degrade structural metrics even when the final result improves. Weight = $W_l = W_{mi}$, half W_b . (<i>methodology.tex:130–133; cunningham1992wycash; kruchten2012technicaldebt; cedrim2017refactoring</i>)	• Lowest solo recovery: 0.132 (inj), 0.014 (us).• Single-knob top-1 disruption = 0.4% (changes top-1 only once).• LOO $\tau = 0.784$ — smallest user-study rank effect.• But : positive-magnitude Ordering DPs rise from 4 $\rightarrow$ 10 of 24 once W_{lag} is included — its real value is in the Ordering detector. (<i>methodology.tex:135; Tables tab:results-ablation-solo, tab:results-sensitivity-knobs</i>)
$W_{cg} \cdot n_{cg}(\tau)$ — long durations without commits	–7	Count of commit-gap events. One event per stretch of ≥ 6 green-refactor checkpoints with no user commit.	Smallest process weight — a deliberately narrow claim. Captures the review/rollback cost of bundled changes, not merge speed (Kudrjavets et al. show no correlation between PR size and time-to-merge across 1.25M PRs). Tao & Kim show separated commits aid review; Di Biase et al. found decomposed changes cut false-positive review comments (1 vs 6, $p = 0.03$); Herzig et al. show tangled commits hurt defect attribution. Weight not fitted — no commit-cadence-to-cost study exists. (<i>methodology.tex:138–142; taokim2015partitioning; dibiasi2019decomposition; herzig2013tangled; kudrjavets2022small</i>)	• Solo recovery = 0.080 (inj), 0.125 (us).• LOO recovery = 0.943 / 0.900; mean LOO $\tau = 0.926$ — affects ranking modestly while leaving most magnitude intact.• Single-knob top-1 disruption = 6.7%. (<i>Tables tab:results-ablation-solo, tab:results-ablation-loo</i>)

Term	Weight	What it measures	Why this term & weight	Stats support
Table 2 — Cleanliness sub-score (6 equally-weighted = 1.0)				
Signal	Tool / aggregation	What it measures	Why this signal	Stats support
Cognitive complexity	SonarSource cognitive complexity, per-method mean	Method-level control-flow comprehension intricacy.	Captures readability/comprehension that pure size metrics miss. Muñoz Barón et al. validate it against 24,400 human ratings (positive correlations with comprehension time + subjective ratings). Counterpoint: Lavazza et al. find it predicts understandability “about as well as size + cyclomatic complexity” — kept as one partial signal in the 6-way composite. (<i>methodology.tex:176–177; campbell2018cognitive; munozbaron2020cognitive; lavazza2023cognitive</i>)	Group-level: cleanliness sub-weights perturb top-1 ranking in at most 0.9% of cases — an order of magnitude less than process weights. (<i>methodology.tex:171; Table tab:results-sensitivity-knobs</i>)
Coupling — CBO	CK library, per-class mean	Number of classes a class depends on.	Classical structural-design signal: lower coupling = better localisation. Foundational metric from Chidamber & Kemerer, widely used as a search-based-refactoring objective. (<i>chidamberkemerer1994ckmetrics; harman2007pareto</i>)	2 top-1 changes / 225 perturbation cases = 0.9% — minimal sensitivity. (<i>Table tab:results-sensitivity-knobs</i>)
Duplication — CPD	PMD CPD, lines on touched files	Lines of code identified as duplicated.	Repeated code = missing abstraction; high maintenance cost (every copy must be updated together). Treated as a design problem + refactoring motivator. (<i>heitlager2007sig; fowler1999refactoring</i>)	No measurable top-1 disruption when its sub-weight is perturbed. Six-part composition “makes the score less dependent on any single metric”. (<i>methodology.tex:147</i>)
Readability	Buse & Weimer feature blend, line-weighted (line length, indentation, identifier characteristics)	Lexical / textual readability — features structural metrics miss.	Combining structural + textual features aligns better with human readability judgements than structural-only models. Caveat: uses the <i>feature taxonomy</i> , not the trained model — calibrating the trained model would need domain-specific data we haven’t collected. (<i>methodology.tex:176; scalabrino2018readability; buse2010readability</i>)	Not isolated in the sensitivity table; bundled into the 6-signal composite. Group-level effect $\leq 0.9\%$ top-1.
Code smells — PMD	PMD violation count on touched files	Rule-based heuristic indicators of possible design problems.	Smells are widely used as refactoring motivators. Foundational theory from Marinescu, Arcelli Fontana et al., Moha et al. Caveat: smell detectors disagree on precision/recall, so PMD is used as one signal among several rather than as ground truth. (<i>background.tex:42–44; marinescu2004; arcellifontana2016falsepositives; moha2010decor; fowler1999refactoring</i>)	Not isolated in the disruption table. Six-metric averaging dilutes single-tool noise — intentional design. (<i>methodology.tex:147</i>)
Cohesion — TCC	CK library, per-class mean; dropped for classes with < 2 methods	Tight Class Cohesion — degree to which class methods use shared state.	High cohesion = well-designed class; low cohesion suggests the class should be split. Complements CBO (inter-class) by capturing intra-class integrity. Widely used in search-based refactoring. (<i>biemankang1995tcc; harman2007pareto</i>)	Not isolated in disruption table. Group-level effect $\leq 0.9\%$ top-1.

Cross-cutting footnotes (memorise these)

- **Top-1 robustness.** Single-knob sweep: **96.5%** of user-study rankable cases keep their top-1 DP under any 1-weight perturbation. Clamp-frozen rate just 1.8% — so almost all of the stability is genuine perturbation response, not a clamp artefact. Multi-knob sweep ($\sigma = \ln 2$, covers $\approx \times 0.25 - \times 4$ of production weights): top-1 drops to **84.6%**, mean Kendall $\tau = 0.586$.
- **Cleanliness vs process weights.** Perturbing any single cleanliness sub-weight changes top-1 in $\leq 0.9\%$ of cases. Process weights change it in 5.3 – 10.7% of cases. $\approx 10\times$ difference — supports the equal-weight cleanliness design.
- **Honest scope limit.** Weights are **not fitted** to outcome data. They follow the literature-severity ordering set in *methodology.tex:80*. Full predictive validation would require an external ground-truth dataset of longitudinal code-quality / maintenance outcomes, which does not currently exist. The score is **defensible locally** — robust to plausible weight perturbations — not claimed to be globally optimal.